MILITARY SPECIFICATION

AMPLIFIER, DOPPLER: 10548313

1. SCOPE

1.1 This specification covers a doppler amplifier assembly, used in Radar Receiver R-1475/VPS-2, part of Radar Set AN/VPS-2.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel; Packaging and Packing of
MIL-I-45607	Inspection Equipment, Acquisition, Maintenance and Disposition of
MIL-C-46357	Computer, Gun Direction; M18 Manufacturing Processes and Inspection

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-810	Environmental Test Methods

DRAWINGS

U. S. Army, Frankford Arsenal

D10548313	Doppler Amplifier

PACKAGING DATA SHEET

10548313	Packaging of Doppler Amplifier

(Copies of specifications, standards, drawings and packaging data sheets required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication. - The doppler amplifier shall be manufactured in accordance with Drawing D10548313 and drawings pertaining thereto.

3.2 General specifications. - Where applicable, the doppler amplifier shall meet the requirements of MIL-F-13926 and MIL-C-46357 as follows:

3.2.1 MIL-F-13926.

a. Order of precedence

b. Dimensions and tolerances

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c. Inorganic protective surface finishings

d. Part identification and marking

e. Workmanship

f. Soldering electrical connections

3.2.2 MIL-C-46357.

a. Etched circuit assemblies

b. Wiring

c. Threaded fasteners

d. Marking

e. Finishes

3.3 Environmental service conditions.

3.3.1 Vibration. - The doppler amplifier shall be mounted on the vibration table in normal service position. The doppler amplifier shall meet the requirements of 3.4 and 3.5 after being subject to continuous vibration for 2 hours along each of three mutually perpendicular axes within the frequency range of 10 to 55 hertz (Hz) and a half amplitude excursion of 1/64 inch. The rate of change of frequency at each frequency range shall not exceed 1 Hz per second. Vibration tests shall be conducted under ambient room conditions.

3.3.2 Shock. - The doppler amplifier shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after being subjected to 18 impact shocks, consisting of three shocks in opposite directions along each of three mutually perpendicular axes. The shock impulses shall have an acceleration of 15 gravity units (G) and a pulse duration of 11 ±1 milliseconds (ms). The G value of each impulse shall be within ±10 percent when measured with a 0.2 to 250 Hz filter and the maximum G shall occur at approximately 5.5 ms.

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3.3.3 Altitude.

3.3.3.1 Operating. - The doppler amplifier shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches Hg).

3.3.3.2 Nonoperating. - The doppler amplifier shall meet the requirements of 3.4 and 3.5 after exposure for not less than 1 hour to barometric pressure encountered at 40,000 feet above sea level (5.5 inches Hg).

3.3.4 Temperature.

3.3.4.1 Operating. - The doppler amplifier shall meet the requirements of 3.4 and 3.5 when stabilized thermally at any temperature between -40° F and +155° F.

3.3.4.2 Nonoperating (storage). - The doppler amplifier shall not suffer permanent damage when subjected to not less than 4 hours at each temperature extreme of -65° F and +160° F, and shall meet the requirements of 3.4 and 3.5 when returned to room ambient.

3.3.5 Humidity (nonoperating). - The doppler amplifier shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity of 95 ±5 percent for a period of not less than 10 days.

3.4 Electrical. - Each conductor shall be electrically continuous as specified on the schematic diagram of the doppler amplifier.

3.5 Performance.

3.5.1 Input voltage. - The input voltage to the doppler amplifier shall be 25.0 ±0.1 volts direct current (Vdc) applied to terminals J1-1 (+) and J1-2 (ret).

3.5.2 Output noise. - With the input signal terminals E1 and E2 open circuited, the output noise at J1-A1 shall be 50 millivolts (mV) peak-to-peak maximum, excluding any 60- Hz noise.

3.5.3 Sensitivity. - With an input signal level of -120 decibel referred to 1 milliwatt (dBm) into 600 ohms sequentially applied to terminals E1 and E2 (signal return) and at the frequencies as specified in Table I, the output levels shall be as specified in Table I.

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TABLE I. - SENSITIVITY LEVELS

FREQUENCY	OUTPUT SIGNAL LEVEL
600 Hz	30.0 mV ±15.0 mV peak-to-peak
5 kHz	35.0 mV ±25.0 mV peak-to-peak
10 kHz	35.0 mV ±25.0 mV peak-to-peak

3.5.4 Limiting action. - With an input signal level of -90 ±5 dBm sequentially applied to terminals E1 and E2 (signal ret) at the frequencies as specified in Table II, the limiting action on the output signal, at J1-A1, shall take palce at the levels as specified in Table II.

TABLE II. - LIMITING ACTION LEVELS

FREQUENCIES	OUTPUT LEVEL AT LIMITING
600 Hz	800 ±130 mV peak-to-peak
5 kHz	700 ±110 mV peak-to-peak
10 kHz	700 ±110 mV peak-to-peak

3.5.5 Stability. - The output signal at J1-A1 shall not break into oscillation throughout an input range of -120 dBm to -30 dBm at frequencies of 600 Hz, 5 kilohertz (kHz), and 10 kHz applied to terminals E1 and E2. (signal ret).

3.5.6 Microphonics test. - The doppler amplifier output signal shall not be severely affected (some distrubance will be noted) or be driven into limiting with the following conditions:

a. The input signal shall be as specified in 3.5.3.

b. The doppler amplifier shall be resting on the test ground plane on the test bench.

c. The test bench shall be given a few short, sharp raps perpendicular to the horizontal axis of the bench.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 General provisions. - The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) submission. - The requirement for first article approval and the responsibility (Government or contractor) for the first article testing shall be as specified in the contract. The sample for first article approval tests shall consit of three doppler amplifiers manufactured in the same manner, using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and production.

4.2.1 Government testing. - When the Government is responsible for the conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to ensure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certification of conformance for materials.

4.2.2 Contractor testing. - When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all requirements of the contract. The sample and a record of this inspection, including certification for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

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4.3 Inspection provisions.

4.3.1 Submission of product. - Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies. - All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provision (SQAP) listed in the technical data package (TDP). In the absence of SQAP's, the applicable quality assurance provisions of MIL-F-13926 shall apply.

4.3.2.2 Final acceptance inspection. - Subsequent to first article approval, examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in 4.3.2.3 and 4.4 herein. Examination and test for packaging and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in 4.3.2.3 and 4.4 shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

4.3.2.3 Functional tests. - The requirements and tests in Table III shall be inspected on a 100 percent basis. All examinations and tests shall be conducted at the standard ambient temperature (60° F and 90° F).

TABLE III. - CLASSIFICATION OF DEFECTS
(100 percent inspection)

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
CRITICAL: NONE		
MAJOR:		
101. Electrical	3.4	4.6.2
102. Performance	3.5	4.6.3
MINOR: NONE		

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4.3.3 Acceptance and rejection. - Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling. - One doppler amplifier shall be selected at random by a Government representative as a control sample from each 100 produced. The control sample shall meet the requirements and tests in Table IV. Three doppler amplifiers shall be selected at random by a Government representative as a control sample from each 50 produced or from each month's production, whichever occurs first. The control samples shall meet the requirements and tests in Table V.

TABLE IV. - SPECIAL SAMPLING

CONTROL TEST	REQUIREMENT	TEST PROCEDURES
301. Fabrication	3.1	Applicable drawings - Visual
302. General specifications	3.2	MIL-F-13926 & MIL-C-46357 - Visual

TABLE V. - SPECIAL SAMPLING, ENVIRONMENTAL

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
303. Vibration	3.3.1	4.6.1.1
304. Shock	3.3.2	4.6.1.2
305. Altitude	3.3.3	4.6.1.3
306. Temperature	3.3.4	4.6.1.4
307. Humidity (nonoperating)	3.3.5	4.6.1.5

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4.4.1 Failure of sample. - Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by Government until necessary corrections have been made by the contractor and a resubmitted sample has been approved (see 4.3.3).

4.5 Inspection equipment. - Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government-furnished inspection equipment. - Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.5.2 Contractor-furnished inspection equipment.

4.5.2.1 Government design. - Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with applicable technical data listed in the TDP.

4.5.2.2 Contractor design. - The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does no exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.2.3 Test equipment. - In conjunction with 4.5.2.2, the standard test equipment listed in Table VI shall be utilized in performing the applicable test of 4.6.

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TABLE VI. - TEST EQUIPMENT LIST

ITEM NUMBER		DESCRIPTION
1.	Audio oscillator	Frequency: 600Hz to 10 kHz. Amplitude: 245 mV. Output impedance: 600 ohms Max. hum & noise: 60 dB below rated output.
2.	ACVM	Voltage range: to 350 mV. Frequency range: 600 Hz to 10 kHz min. Accuracy: ±1%. Input impedance: 10 megohms.
3.	Audio variable attenuator	Attenuation: 20 dB to 110 dB in 10-dB steps. Accuracy: ±0.5 dB max. Impedance: 600 ohms.
4.	Power supply	Output voltage: +25.0 ±0.1 Vdc. Load regulation: 0.01% max. Line regulation: 0.01% max. Ripple & noise: 200 μV max. Output current: 200 mA min.
5.	Oscilloscope	Bandwidth: 1 MHz min. Deflection factor: ac 10 mV/div to 1 V/div. Input impedance: 1 megohm shunted by approximately 40 pF.
6.	Brass test plate	2 ft by 3 ft, approx.
7.	Termination	150 ohm, 1/2 W.
8.	Connector	DEM-5WIS, Cinch Mfg. Co., Chicago, Ill., or equivalent.

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4.6 Test methods and procedures.

4.6.1 Environmental. - Environmental tests shall be conducted in accordance with the general requirements and test methods of MIL-STD-810 except as specified herein. Testing equipment used in environmental tests shall be in accordance with "Test Facilities and Apparatus" requirements of MIL-STD-810.

4.6.1.1 Vibration. - The test shall be performed as specified in Method 514, Procedure IX, Part I of MIL-STD-810 except that failure criteria do not apply. After vibration, the doppler amplifier shall be tested as specified in 4.6.2 and 4.6.3 to determine compliance with 3.3.1.

4.6.1.2 Shock. - This test shall be performed as specified in Procedure I, Method 516 of MIL-STD-810 except that shock pulses shall be 11.0 ms and 15 G. After the shock impulses, the doppler amplifier shall be tested as specified in 4.6.2 and 4.6.3 to determine compliance with 3.3.2.

4.6.1.3 Altitude (operating and nonoperating). - This test shall be performed as specified in Procedure I, Method 500 of MIL-STD-810 except that in Step 2 the altitude shall be limited to 40,000 feet (5.5 in Hg). Operation of Step 3 shall be the performance tests of 4.6.3.

4.6.1.4 Temperature.

4.6.1.4.1 High (operating and nonoperating). - This test shall be performed as specified in Procedure I, Method 501 of MIL-STD-810. The temperature of Step 2 shall be +160° F. The temperature of Step 3 shall be maintained for not less than 4 hours. The temperature of Step 4 shall be +155° F. In Step 5, the temperature shall be maintained for 4 hours and the doppler amplifier shall be tested in accordance with 4.6.2 and 4.6.3 during this time.

4.6.1.4.2 Low (operating and nonoperating). - This test shall be performed as specified in Procedure I, Method 502 of MIL-STD-810. The temperature of Step 2 shall be -65° F maintained for not less than 4 hours. The temperature of Step 4 shall be -40° F. In Steps 5 and 7, the doppler amplifier shall be tested in accordance with 4.6.2 and 4.6.3.

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4.6.1.5 Humidity (nonoperating). - This test shall be performed in accordance with Procedure II, Method 507 of MIL-STD-810. The doppler amplifier shall be tested in accordance with 4.6.3.

4.6.2 Electrical. - Continuity shall be inspected by standard measuring equipment (SME), and visual examination for compliance with 3.4.

4.6.3 Performance. - The selected doppler amplifier shall be tested as follows: Set up the equipment as shown in Figure 1. Do not apply power to the doppler amplifier under test. All test equipment is to be connected to its 110-V, 60-Hz source through three-prong grounding plugs. All test equipment, except the oscilloscope, shall be placed on the brass test plate, item 6. The oscilloscope must have a ground strap connected to the brass test plate.

4.6.3.1 Input voltage. - Apply the input voltage as specified in 3.5.1.

4.6.3.2 Output noise. - With no input signal applied to terminals E1 and E2, use the oscilloscope, item 5, to measure the output noise level as specified in 3.5.2.

4.6.3.3 Sensitivity. - Adjust the attenuator, item 3, for 110 decibels (dB) and connect the high side output to E1 and the low side to E2 of the doppler amplifier. Adjust the audio oscillator, item 1, for an output level of 245 mV as read on the alternating current (ac) voltmeter, item 2 (this equals the -120 dBm input into 600 ohms as specified in 3.5.3). Sequentially apply this level at the frequencies as specified in Table I and observe the output signal level at J1-A1 to be as specified in Table I.

4.6.3.4 Limiting action. - Use the oscilloscope to observe the top of one cycle of the output signal and decrease the variable attenuator setting, item 3, until the top of the sine wave signal starts to flatten out. The reading on the attenuator with 10 dB added to it shall be the input signal level as specified in 3.5.4. The output signal at the point of limiting action shall be as specified in Table II.

4.6.3.5 Stability. - The output signal observed at J1-A1 shall be as specified in 3.5.5.

4.6.3.6 Microphonics test. - With the conditions of 3.5.6 the doppler amplifier shall not be severely affected (some disturbance will be noted) or be driven into limiting.

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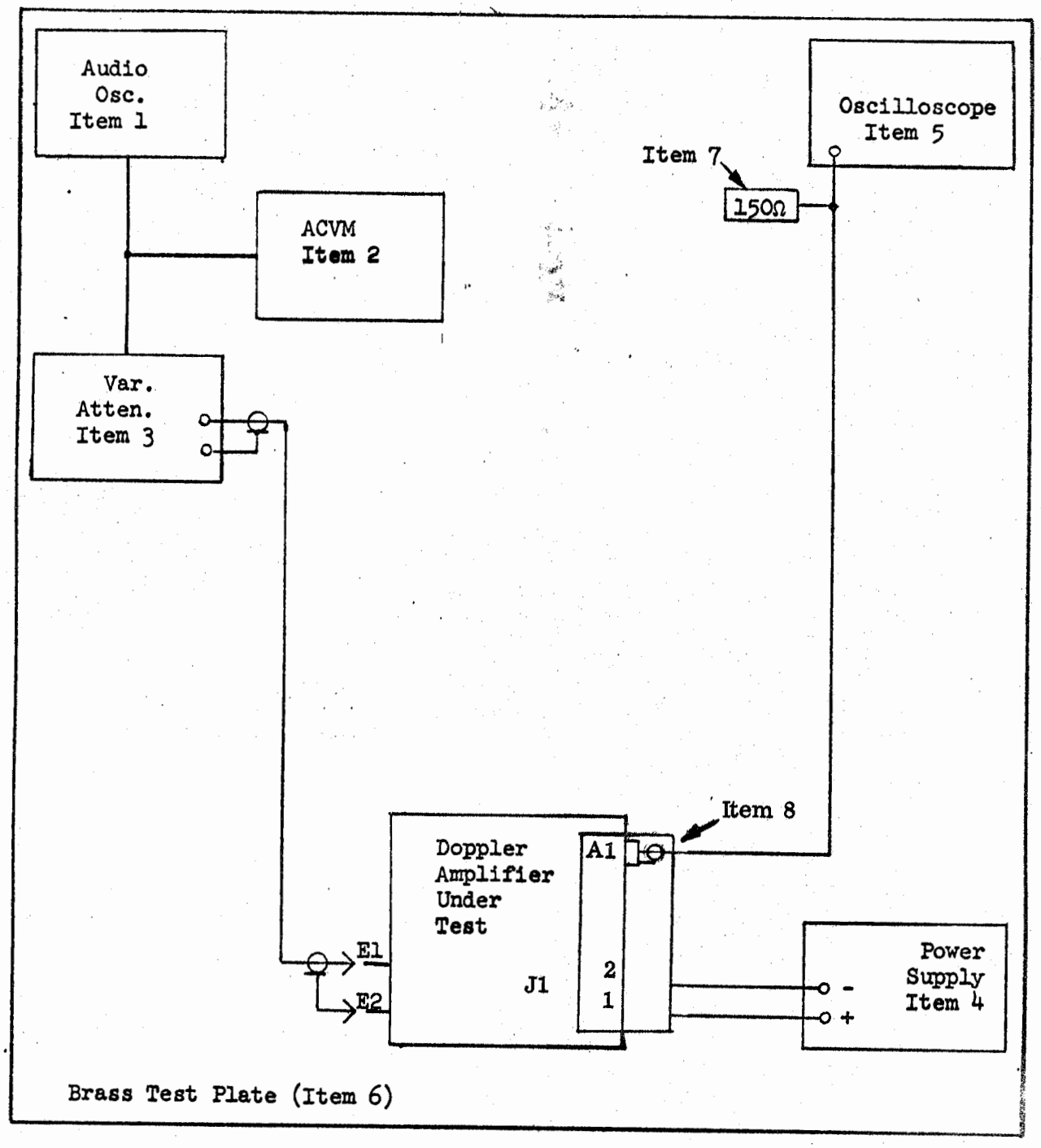


FIGURE 1. TEST EQUIPMENT SETUP.

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5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. - Packaging, packing, and marking shall be in accordance with Packaging Data Sheet 10548313. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. - The doppler amplifier is used in Radar Receiver R-1475/VPS-2, part of Radar Set AN/VPS-2. It is a broadband amplifier which provides amplification of low-level doppler signals and supplies then to a phase detector.

6.2 Ordering data. - Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Applicable packaging data sheet number (see 5.1).

c. Selection of applicable level of preservation packaging and packing.

d. Level of protection required.

e. Applicable stock number.

f. Requirement for first article submission.

Custodian:

Army - MU

Preparing Activity:

Army - MU

Project No. 1285-A285

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